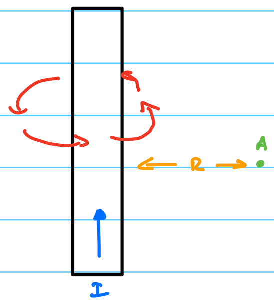
Electromagnetism Basics



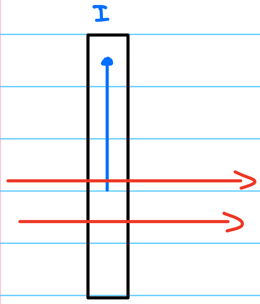
Magnetic Field Strength

$$B = \frac{\mu_0 I}{2\pi R}$$ Units in Tesla

$\mu_0 = 4\pi \times 10^{-7}\ T.m/A$ → Permeability of free space

$I\uparrow \quad B\uparrow$

$R\uparrow \quad B\downarrow$



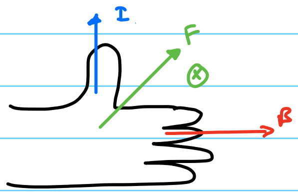
Magnetic Force

$$\boxed{F = ILB\sin\theta}$$

$I\uparrow \quad F\uparrow \qquad L\uparrow \quad F\uparrow$

$B\uparrow \quad F\uparrow$

Right Hand Rule



$I = \frac{Q}{t} = \frac{qN}{t}$

$d = Vt \quad \therefore L = Vt$

$\therefore F = \frac{qN}{\cancel{t}} \cdot V\cancel{t}\, B\sin\theta$

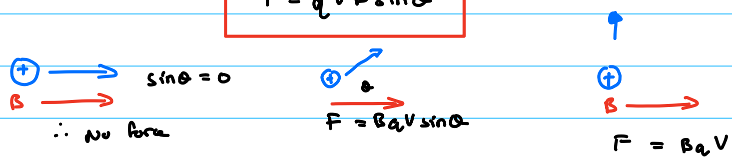
$F = qNVB\sin\theta$

$N = 1$

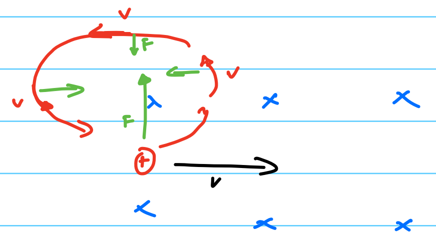
$F = qVB\sin\theta$

Force of a charged particle

$$\boxed{F = qVB\sin\theta}$$



Magnetic Centripetal force

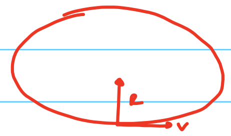


$F_c = F_B$

$\frac{mv^2}{R} = Bqv$

$\therefore mv = BqR$

$\boxed{mv = BqR}$

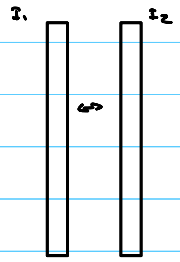


Magnetic Field between Two wires

$B_1 = \frac{\mu_0 I_1}{2\pi R}$

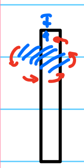
$F_2 = I_2 L B_1$

$\boxed{F_2 = \frac{I_2 L \mu_0 I_1}{2\pi R}}$



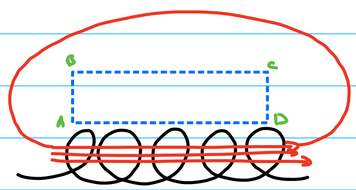
Ampere's Law

$\sum B_{\parallel} \Delta L = \mu_0 I_{enclosed}$



$B = \frac{\mu_0 I}{2\pi R}$

• Ampere's Law: Solenoid



$\sum B_{\parallel} \Delta l = (Bl)_{AC} + \cancel{(Bl)_{CD}}^{0} + \cancel{(Bl)_{DB}}^{0} + \cancel{(Bl)_{BA}}^{0}$

CD ⊥ AC　　weak　　BA ⊥ AC

$\therefore Bl = \mu_0 I_{en}$ for 1 loop

$\therefore Bl = \mu_0 IN$

magnetic Field of a Solenoid

$B = \frac{\mu_0 N I}{l}$

$n = \frac{N}{l}$ → loops, → meters

Turns / meter

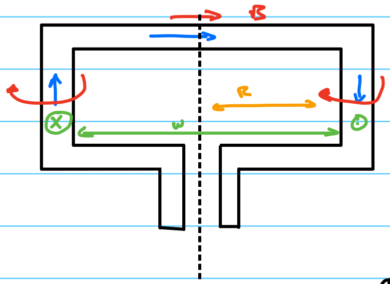
$\boxed{B = \mu_0 n I}$

n↑ B↑

L↓ B↑

I↑ B↑

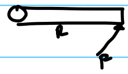
Torque



$F = ILB$

$\tau = \tau_1 + \tau_2$

$\tau = FR$



$\tau = FR\sin\theta$

$R = \frac{w}{2}$

$F = ILB$

$\therefore \tau = ILB\left(\frac{w}{2}\right)\sin\theta + ILB\left(\frac{w}{2}\right)\sin\theta$

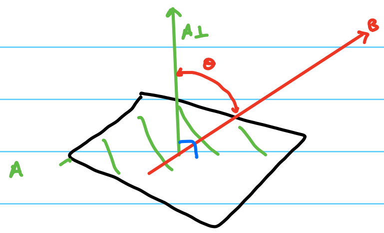
$\tau = ILB\sin\theta\left(\frac{w}{2} + \frac{w}{2}\right)$

$\tau = I\underbrace{Lw}_{A}B\sin\theta$

$\tau = IAB\sin\theta$

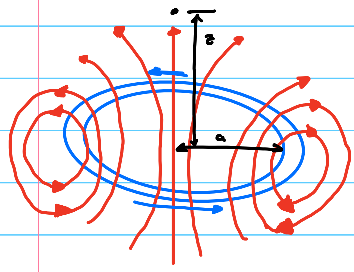
$\tau = NIAB\sin\theta$

no. loops



$\boxed{\tau = NIAB\sin\theta}$

Field of a Magnetic Dipole



$$B_z = \frac{\mu_0 I a^2}{2(z^2+a^2)^{3/2}}$$

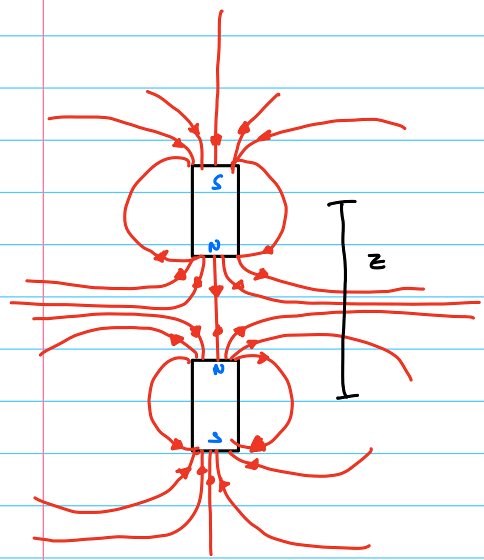
$$= \frac{\mu_0 I A}{2\pi(z^2+a^2)^{3/2}} \qquad m = Ia$$

$$= \frac{\mu_0 m}{2\pi(z^2+a^2)^{3/2}} \qquad \boxed{B_z = \frac{\mu_0 m}{2\pi z^3}}$$

if $z \gg a$

$$\therefore \quad B_z = \frac{\mu_0 m}{2\pi z^3} = \frac{\mu_0 m}{2\pi} \cdot \frac{1}{z^3}$$

Force of Dipole magnet on another dipole



Energy of a dipole in a magnetic field

$$U = -mB\cos\theta$$

The Force Between Dipoles

θ = angle between field & normal vector of area of current loop

As a particular example of the force (3.28), consider the case where ... is set up by a dipole $\mathbf{m}_1$. We know that the resulting long-distan... (3.24),

$\therefore$ aligned ($\theta = 0$)

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi}\left(\frac{3(\mathbf{m}_1\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}_1}{r^3}\right)$$

anti-aligned ($\theta = 180°$)

Now we'll consider how this affects the second dipole $\mathbf{m} = \mathbf{m}_2$. Fro...

$$\mathbf{F} = \frac{\mu_0}{4\pi}\nabla\left(\frac{3(\mathbf{m}_1\cdot\hat{\mathbf{r}})(\mathbf{m}_2\cdot\hat{\mathbf{r}}) - \mathbf{m}_1\cdot\mathbf{m}_2}{r^3}\right)$$

where $\mathbf{r}$ is the vector from $\mathbf{m}_1$ to $\mathbf{m}_2$. Note that the structure of t... to that between two electric dipoles in (2.30). This is particular... we used two rather different methods to calculate these forces. ... derivative, we have

$$\mathbf{F} = \frac{3\mu_0}{4\pi r^4}\left[(\mathbf{m}_1\cdot\hat{\mathbf{r}})\mathbf{m}_2 + (\mathbf{m}_2\cdot\hat{\mathbf{r}})\mathbf{m}_1 + (\mathbf{m}_1\cdot\mathbf{m}_2)\hat{\mathbf{r}} - 5(\mathbf{m}_1\cdot\hat{\mathbf{r}} ...\right.$$

$$\therefore \quad F_z = -\frac{dU}{dz}$$

Aligned Poles:

$$F_z = -\frac{m_s \mu_0 m_m}{2\pi} \cdot \frac{d}{dz}\frac{1}{z^3}$$

$$F_z = \frac{3 m_s \mu_0 m_m}{2\pi} \cdot \frac{1}{z^4}$$

m_s = source magnetic dipole $\quad$ m_m = feeling it.

Anti-Aligned:

$$F_z = -\frac{3 m_s \mu_0 m_m}{2\pi} \cdot \frac{1}{z^4}$$

First note that if we swap $\mathbf{m}_1$ and $\mathbf{m}_2$, so that we also send $\mathbf{r} \to -\mathbf{r}$, then the force swaps sign. This is a manifestation of Newton's third law: every action has an equal and opposite reaction. Recall from Dynamics and Relativity lectures that we needed Newton's third law to prove the conservation of momentum of a collection of particles. We see that this holds for a bunch of dipoles in a magnetic field.

But there was also a second part to Newton's third law: to prove the conservation of angular momentum of a collection of particles, we needed the force to lie parallel to the separation of the two particles. And this is *not* true for the force (3.31). If you set up a collection of dipoles, they will start spinning, seemingly in contradiction of the conservation of angular momentum. What's going on?! Well, angular momentum is conserved, but you have to look elsewhere to see it. The angular momentum carried by the dipoles is compensated by the angular momentum carried by the magnetic field itself.

Finally, a few basic comments: the dipole force drops off as $1/r^4$, quicker than the Coulomb force. Correspondingly, it grows quicker than the Coulomb force at short distances. If $\mathbf{m}_1$ and $\mathbf{m}_2$ point in the same direction and lie parallel to the separation $\mathbf{R}$, then the force is attractive. If $\mathbf{m}_1$ and $\mathbf{m}_2$ point in opposite directions and lie parallel to the separation between them, then the force is repulsive. The expression (3.31) tells us the general result.

Magnetic Dipole Moment

- Vector Quantity
- Magnetic Strength & Orientation

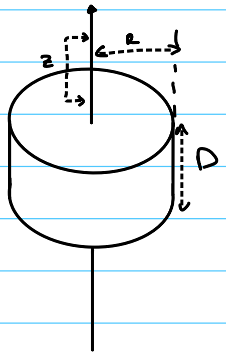
$$m = I\,A$$

current, Area

remenance, Volume

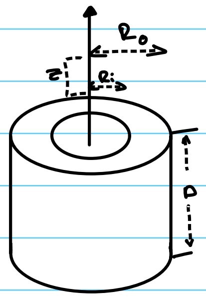
$$m = \frac{B_r\,V}{\mu_0}$$

constant

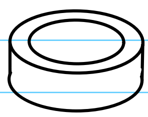


Magnetic Flux Density of a Cylinder

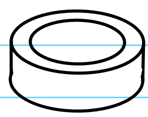
$$B = \frac{Br}{2}\left(\frac{D+z}{\sqrt{R^2+(D+z)^2}} - \frac{z}{\sqrt{R^2+z^2}}\right)$$



$$B = \frac{Br}{2}\left[\frac{D+z}{\sqrt{R_a^2+(D+z)^2}} - \frac{z}{\sqrt{R_a^2+z^2}} - \left(\frac{D+z}{\sqrt{R_i^2+(D+z)^2}} - \frac{z}{\sqrt{R_i^2+z^2}}\right)\right]$$



OD = 32mm
ID = 18mm
t = 5.5mm

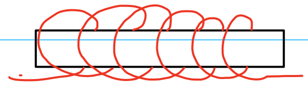


OD = 50mm
ID = 12mm
t = 8mm

$$F = \frac{B^2 A}{2\mu_0}$$

$\mu_0 = 4\pi \times 10^{-7}$

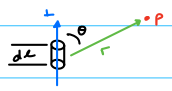
Electromagnet Design

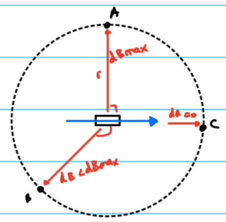


$B = \mu_0 \frac{NI}{l}$ $\qquad$ $F = \frac{B^2 S}{2\mu_0}$ $\rightarrow$ coil cross section area

Electromagnetic Fields

Biot-Savart's Law





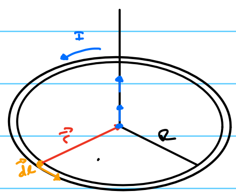
$$\vec{dB} = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot \overbrace{\vec{dl} \times \hat{r}}^{\text{cross product}}}{r^2}$$

$\mu_0 : 4\pi \times 10^{-7}$ (permeability of vacuum)

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I\, dl \sin\theta}{r^2}$$

$$\therefore \vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{\vec{dl} \times \vec{r}}{r^3}$$

Example: Circular loop at center



$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{\vec{dl} \times \vec{r}}{r^3}$ $\qquad$ $B = \frac{\mu_0 I}{4\pi} \int \frac{\vec{dl} \times \hat{r}}{r^2}$ $\qquad$ $(\vec{r} = r\hat{r})$

$B = \int dB$

$dB = \frac{\mu_0 I}{4\pi} \frac{dl\, r \sin\theta}{r^3}$

$\theta = \frac{\pi}{2} \quad \therefore \sin(\theta) = 1$

$\therefore dB = \frac{\mu_0}{4\pi} \cdot \frac{dl \cdot r}{r^{\cancel{3}\,2}}$

$B = \frac{\mu_0 I}{4\pi} \int \frac{dl}{r^2}$

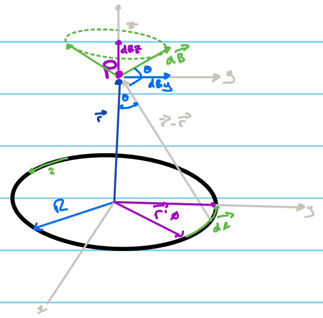
$B = \frac{\mu_0 I}{4\pi r^2} \int dl \rightarrow \int dl = 2\pi R$

$= \frac{\mu_0 I}{4\pi R^2} \cdot 2\pi R$

$B = \frac{\mu_0 I}{2R}$

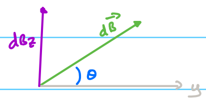
$$\boxed{\vec{B} = \left(\frac{\mu_0 I}{2R}\right)\hat{k}}$$

Biot - Savart Law: Circular Loop @ Point



$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{I\, d\vec{l} \times (\vec{r}-\vec{r}')}{\|\vec{r}-\vec{r}'\|^3}$$

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I\, dl\, \|r-r'\|}{\|r-r'\|^3} = \frac{\mu_0}{4\pi} \cdot \frac{I\, dl}{\|\vec{r}-\vec{r}'\|^2}$$



$$\|\vec{r}-\vec{r}'\|^2 = R^2 + z^2 \qquad dl = R\, d\phi$$

$$\therefore \quad dB = \frac{\mu_0}{4\pi} \cdot \frac{I\, dl}{(R^2+z^2)}$$

$$\therefore \quad dB = \frac{\mu_0}{4\pi} \cdot \frac{I R\, d\phi}{(R^2+z^2)}$$

$$\sin\theta = \frac{R}{(R^2+z^2)^{\frac{1}{2}}}$$

$$\therefore \quad dB_z = dB \sin\theta = \frac{\mu_0}{4\pi} \cdot \frac{I R\, d\phi}{(R^2+z^2)} \cdot \left[\frac{R}{(R^2+z^2)^{\frac{1}{2}}}\right]$$

$$B_{z,total} = \int_0^{2\pi} dB_z = \frac{\mu_0}{\cancel{4\pi}\ 2\pi} \cdot \frac{I R^2}{(R^2+z^2)^{3/2}} \underbrace{\int_0^{2\pi} d\phi}_{2\pi}$$

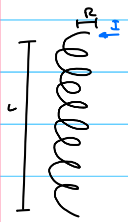
$$B_{z,total} = \frac{\mu_0 I R^2}{2(R^2+z^2)^{3/2}}$$

$$\boxed{B_z = \frac{\mu_0}{2} \cdot \frac{I R^2}{(R^2+z^2)^{\frac{3}{2}}}}$$

Magnetic Field Strength: Solenoid

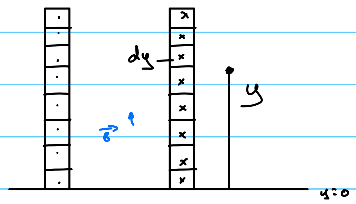
Assumptions

- Tightly wound
- Thickness of coil constant (uniform coil density)
- coil density constant



I_T = total current running throug cross section

that encompasses all coils

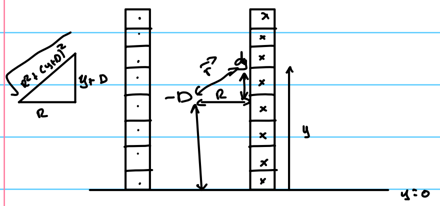


N = # coils

n = coil density $= \frac{N}{L} = \frac{dN}{dy}$

$$\therefore \quad I_T = NI = nLI = nIL$$

$$dI_T = nI\, dy = \frac{NI}{L}\, dy$$



D = distance from end of solenoid to point on a central axis

$\therefore$ $D<0$ = inside solenoid

$D>0$ = outside

If we consider loop of width dy & point $y+D$

$$d\vec{B} = \frac{\mu_0 R^2 \, dI_T}{2(R^2+(y+D)^2)^{3/2}} \hat{j}$$

$$d\vec{B} = \frac{\mu_0 R^2}{2}\hat{j} \cdot \frac{\frac{NI}{L}dy}{(R^2+(y+D)^2)^{3/2}} = \hat{j}\,\frac{\mu_0 R^2}{2L} \cdot \frac{dy}{(R^2+(y+D)^2)^{3/2}}$$

let $u = y+D$ $\therefore du = dy$ $\quad \int_0^L dy \Rightarrow \int_D^{L+D} du$

$$\vec{B} = \frac{\mu_0 R^2 NI}{2L} \cdot \int_D^{L+D} \frac{du}{(R^2+u^2)^{3/2}} = \hat{j}\,\frac{\mu_0 R^2 NI}{2L}\left[\frac{u}{R^2\sqrt{R^2+u^2}}\right]_D^{L+D}$$

$$\vec{B} = \hat{j}\left[\frac{L+D}{\sqrt{R^2+(L+D)^2}} - \frac{D}{\sqrt{R^2+D^2}}\right]$$

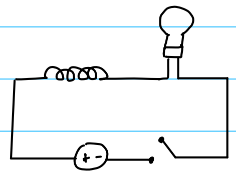
$$\boxed{B(z) = \frac{\mu_0 n I}{2}\left(\frac{z+\frac{L}{2}}{\sqrt{R^2+(z+\frac{L}{2})^2}} - \frac{z-\frac{L}{2}}{\sqrt{R^2+(z-\frac{L}{2})^2}}\right)}$$

$$\frac{dB_z}{dz} = \frac{\mu_0 n I R^2}{2}\left[\frac{1}{(R^2+(z-\frac{L}{2})^2)^{\frac{3}{2}}} - \frac{1}{(R^2+(z+\frac{L}{2})^2)^{3/2}}\right]$$

$\therefore$

$$\boxed{\begin{aligned} F_z &= m \cdot \frac{dB}{dz} \\ F &= m \cdot \frac{\mu_0 n I R^2}{2}\left[\frac{1}{(R^2+(z-\frac{L}{2})^2)^{\frac{3}{2}}} - \frac{1}{(R^2+(z+\frac{L}{2})^2)^{\frac{3}{2}}}\right] \end{aligned}}$$

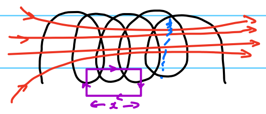
Inductance



$$\varepsilon = -N\frac{d\phi_B}{dt} = -L\frac{di}{dt}$$

change in flux

$$\varepsilon = -L\frac{di}{dt}$$



$$N\phi_B = Li$$

$$\therefore\ L = \frac{N\phi_B}{i}$$

Ampere's Law

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$$

$$B\cdot\cancel{x} = \frac{\mu_0 N \cancel{x} i}{l}$$

$$L = \frac{N(BA)}{i} \quad \therefore\ L = \frac{N\left(\frac{\mu_0 N \cancel{i}}{l}\right)A}{\cancel{i}}$$

$$L = \frac{\mu_0 N^2 A}{l}$$

$$\boxed{L = \frac{\mu_0 N^2 A}{l}}$$